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Quantum Computing

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Complex linear algebra

1.1 Complex numbers

Complex number

A **Complex number** z is defined as:

$$z = a + ib$$

where $b \in \mathbb{R}$.

From a complex number we can define:

- Coniugate: $\bar{z} = \overline{a + ib} = a - ib$
- Modulus: $|z| = \sqrt{a^2 + b^2}$

Given a number z and its modulus $|z|$, we can find an angle $\theta \in [-\pi, \pi]$ such that:

$$\begin{aligned}\cos \theta &= \frac{a}{\sqrt{a^2 + b^2}} = \frac{a}{|z|} \\ \sin \theta &= \frac{b}{\sqrt{a^2 + b^2}} = \frac{b}{|z|}\end{aligned}$$

So, we can write:

$$z = |z|(\cos \theta + i \sin \theta)$$

We also define the exponential of an imaginary number:

$$e^{i\theta} = \cos \theta + i \sin \theta \implies z = |z|e^{i\theta}$$

1.2 Vectors and Matrices

A vector n is list of complex numbers, for which we can define a norm:

$$||v|| = \sqrt{|v_1|^2 + \dots + |v_n|^2} = \sqrt{\sum_{i=1}^n |v_i|^2}$$

Example:

Given the vector:

$$v = [4, i, 2 + 3i]$$

Its norm is:

$$||v|| = \sqrt{16 + 1 + 13} = \sqrt{30}$$

A matrix is a table of complex number, for which the multiplication is not commutative. The multiplication between a matrix A with size $m \times n$ and a matrix B with size $n \times q$ has size $m \times q$. The result matrix C has in every cell:

$$C_{i,j} = \sum_{k=0}^n A_{i,k} \cdot B_{k,j}$$

Example:

Given the matrices:

$$A = \begin{bmatrix} 3 & i \\ 1+i & 0 \\ -4 & 2+3i \end{bmatrix} \quad B = \begin{bmatrix} 3 & 1 \\ 0 & 0 \end{bmatrix}$$

The multiplication $A \cdot B$ is:

$$C = \begin{bmatrix} 9 & 3 \\ 3+3i & 1+i \\ -12 & -4 \end{bmatrix}$$

1.2.1 Eigenvectors and Eigenvalues

A vector v is an **eigenvector** with **eigenvalue** λ for a matrix A if:

$$A \cdot v = \lambda \cdot v$$

We can also write:

$$A \cdot v - \lambda \cdot v = 0 \iff (A - \lambda I) \cdot v = 0$$

This equation has non-zero solutions only if:

$$\det(A - \lambda I) = 0$$

In general v and λ could be complex, but quantum mechanics uses matrices that always have eigenvalues $\lambda \in \mathbb{R}$.

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Introduction

2.1 Probabilistic algorithms

In normal programming, an algorithms produce a fixed output for a given input. With probabilistic algorithms there is a finite set of possible outputs $\{o_1, \dots, o_N\}$ for a given input. Each outputs o_i has a probability of p_i to be produced, where:

$$\sum_{k=1}^N p_k = 1$$

Classical algorithms can have probabilistic outputs based on pseudo-random number generator (PRNG), that produce a sequence of numbers that appear random but after many elements the sequence repeats. In quantum algorithms there is no PRNG, but real randomness.

2.2 Qubit

Qubit

A **qubit** is the quantum equivalent of a bit, where the basis states are:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The true state of a qubit ψ is a **superposition** defined as:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

where $\alpha, \beta \in \mathbb{C}$ are called **probability amplitudes** and $|\alpha|^2 + |\beta|^2 = 1$.

The notation $|\psi\rangle$ is called "ket".

Example:

Given the vector:

$$v = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

The corresponding qubit is:

$$\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

The true state of a qubit cannot be observed, but only measured. Measuring a qubit makes it collapse into $|0\rangle$ or $|1\rangle$ with the probabilities $|\alpha|^2$ and $|\beta|^2$. We write this property as:

$$\alpha |0\rangle + \beta |1\rangle \begin{cases} |0\rangle & @ |\alpha|^2 \\ |1\rangle & @ |\beta|^2 \end{cases}$$

where @ is called "at".

For any complex number z such that $|z| = 1$, we have that $|z\alpha| = |\alpha|$, so when measuring we have that:

$$z\alpha |0\rangle + z\beta |1\rangle = \alpha |0\rangle + \beta |1\rangle$$

because the probability is based on the amplitudes squared.

2.3 Qubit operations

Qubits transformations are linear, so they can be represented as a matrix.

Pauli matrices

The **Pauli matrices** are 2×2 matrices that represent operations:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

These are called the X , Y and Z gate respectively.

The result of applying this gates on the two basis states $|0\rangle$ and $|1\rangle$ are:

- X gate (also called NOT):

$$\sigma_x |0\rangle = |1\rangle \quad \sigma_x |1\rangle = |0\rangle$$

- Y gate:

$$\sigma_y |0\rangle = i |1\rangle \quad \sigma_y |1\rangle = -i |0\rangle$$

- Z gate:

$$\sigma_z |0\rangle = |0\rangle \quad \sigma_z |1\rangle = -|1\rangle$$

Another important transformation is the **Hadamard gate**, represented by the following matrix:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

This gate can map a classical state into a superposition:

$$H |0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

If we measure it we will have a true random bits generator:

$$\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \begin{cases} |0\rangle & @ \frac{1}{2} \\ |1\rangle & @ \frac{1}{2} \end{cases}$$

To be an allowed quantum transformation a matrix must preserve the norm of the input vector, so it must be **unitary**. Since $|\alpha|^2 + |\beta|^2 = 1$, the norm of every vector v must be $\|v\| = 1$ for any qubit state. A unitary matrix preserve the probabilities and is reversible, so from the output we can recover the input.

2.3.1 Bloch sphere

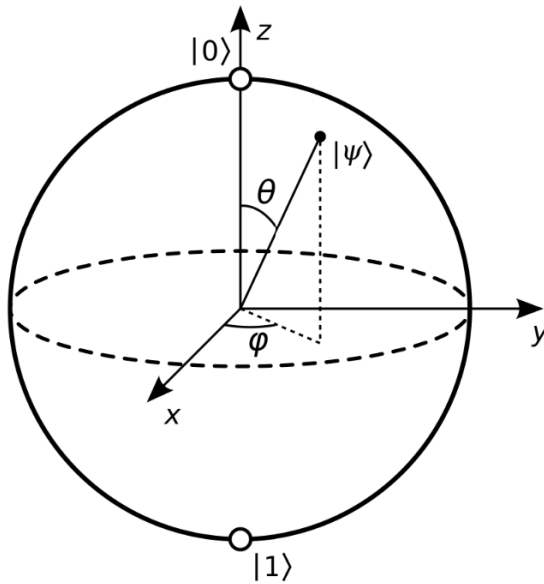
When can picture a qubit using the **Bloch sphere**. Considering a qubit:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

Rewriting it using polar coordinates:

$$|\psi\rangle = e^{i\gamma} \left(\cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle \right)$$

Since $|e^{i\gamma}| = 1$ we can remove it, representing the qubit based on two angles $\theta \in [0, \pi]$ and $\varphi \in [0, 2\pi)$.



There is no Bloch sphere equivalent for systems composed by two or more qubits.